Abstract

Objective: To compare the effects of a tight glycated hemoglobin (HbA1c) target (< 7.0%) versus a less tight target (< 7.5%) on microvascular complications in adults with poorly controlled type 2 diabetes (T2D) in Egypt.

Material and methods: In this parallel-group, randomized controlled trial, 80 adults (18–60 years) with T2D, baseline HbA1c > 7.5%, and mild-to-moderate non-proliferative diabetic retinopathy (NPDR) were randomized by computer-generated simple randomization to a tight target (Group A; n = 33) or a less tight target (Group B; n = 47) and followed for 6 months. Glycemic indices, renal parameters (albumin-to-creatinine ratio and estimated glomerular filtration rate), and neuropathy (Douleur Neuropathique 4 questionnaire) were assessed at baseline and 6 months. Retinopathy worsening was defined as ≥ 1-step progression on the Early Treatment Diabetic Retinopathy Study (ETDRS) severity scale.

Results: At 6 months, Group A achieved lower HbA1c than Group B [6.23% ± 0.35 (45 mmol/mol) vs. 7.27% ± 0.23 (56 mmol/mol); $p < 0.001$], with lower fasting plasma glucose and 2-hour postprandial glucose (both $p < 0.001$). Retinopathy worsening occurred more frequently in Group A than Group B (63.6% vs. 14.9%; $p < 0.001$). Overall new-onset or progressive microvascular complications were higher in Group A (84.8%) than Group B (31.9%; $p < 0.001$), including albuminuria (36.4% vs. 14.9%; $p = 0.026$) and neuropathy (39.4% vs. 10.6%; $p = 0.002$).

Conclusions: Although tighter glycemic targets improved glycemic indices, they were associated with substantially higher short-term retinopathy worsening and greater overall microvascular deterioration. Individualized targets and gradual HbA1c reduction with close ophthalmologic follow-up may mitigate early worsening risk. Trial registration: ClinicalTrials.gov NCT07049601.

Keywords: type 2 diabetes, tight glycemic control, diabetic retinopathy, microvascular complications, randomized controlled trial

Introduction

Type 2 diabetes mellitus (T2D) is marked by chronic hyperglycemia from insulin resistance and impaired insulin secretion. Its high prevalence and associated vascular complications make it a major global health burden, with microvascular complications — especially neuropathy, nephropathy, and retinopathy — significantly increasing morbidity and reducing quality of life [1–3].

Glycemic control, commonly assessed using glycated hemoglobin (HbA1c), is central to diabetes management. Clinical trials such as Diabetes Control and Complications Trial (DCCT) and United Kingdom Prospective Diabetes Study (UKPDS) have shown that tight glycemic control reduces incidence and slows microvascular complications progression [2, 4, 5]. The (UKPDS) further demonstrated that improved glycemic control confers long-term "legacy effects," with sustained reductions in microvascular complications observed years after initial intervention [6]. This evidence supports early optimization of HbA1c to mitigate long-term vascular risk.

However, intensive lowering can be associated with harm, including hypoglycemia and the phenomenon of early worsening of diabetic retinopathy following rapid improvement in glycemia [1, 7, 8].

Importantly, early worsening has been linked to both the magnitude of HbA1c reduction and the rate at which HbA1c falls. Disentangling these effects typically requires frequent interim HbA1c measurements and analytic modeling of HbA1c change over time; therefore, a target-based comparison over a fixed follow-up primarily reflects achieved glycemic intensity rather than formally isolating “rate” versus “magnitude” as independent contributors [9].

In addition, glycemic exposure is not fully captured by HbA1c alone. Emerging data suggest that glycemic variability and postprandial excursions may contribute to microvascular risk and could help explain heterogeneity in outcomes despite similar HbA1c values. Nonetheless, HbA1c remains the principal treatment target recommended by guidelines and widely used in routine practice, making it a pragmatic anchor for interventional comparisons.

The American Diabetes Association (ADA) Standards of Care 2024 recommend individualized glycemic targets, with HbA1c < 7% for most adults, adjusted according to comorbidities, complication burden, and hypoglycemia risk [10]. Despite guidelinerecommended individualized targets, the optimal extent and speed of HbA1c reduction remain unclear, especially in patients with microvascular disease, as tight control may not benefit all and can cause harm in some subgroups [11].

Therefore, this study aimed to compare the effects of tight glycemic control (HbA1c < 7.0%) with less tight control (HbA1c < 7.5%) on microvascular complications in Egyptian adults with poorly controlled T2D.

Material and methods

Participants

Eighty adults with inadequately controlled T2D were recruited. Eligibility required a diagnosis of T2D according to ADA 2024 criteria [10], including HbA1c-based diagnostic thresholds and individualized glycemic targets based on risk profile and complication status. Participants were 18–60 years of age and had mild-to-moderate non-proliferative diabetic retinopathy (NPDR) at baseline.

Participants were excluded if they had type 1 diabetes or advanced diabetic retinopathy. Additional exclusion criteria were ESRD [estimated glomerular filtration rate (eGFR) < 15 mL/min/1.73 m^2], chronic liver disease, malignancy, or a recent myocardial infarction. Patients with frequent hypoglycemia or hypoglycemia unawareness were excluded because intensification of therapy could increase short-term risk.

Sample size calculation

Sample size was estimated using diabetic retinopathy progression rates reported in Diabetes Control and Complications Trial (DCCT). A minimum of 33 participants per group was required to detect a statistically significant difference with 80% power at an alpha level of 0.05. To account for potential attrition, 80 participants were enrolled.

The sample size calculation was based on progression rates derived from a type 1 diabetes population. This approach was used as a conservative estimate, but those rates may not fully reflect progression risk in individuals with T2D.

Randomization and interventions

Randomization was performed using a computer-generated simple random allocation sequence created with Random Allocation Software (version 2.0). Because simple randomization was used without blocking or stratification, group sizes were unequal (33 in Group A and 47 in Group B). Allocation concealment was ensured using sequentially numbered, opaque, sealed envelopes prepared by an independent statistician who was not involved in patient enrollment or outcome assessment.

Because glycemic targets differed between groups, participants and treating clinicians were not blinded. To reduce detection bias, outcome assessors were blinded to assignment, including ophthalmologists evaluating fundus images, laboratory staff analyzing biochemical markers, and investigators assessing neuropathy.

Participants were assigned to one of two parallel groups. Group A followed a tight glycemic control strategy with a target HbA1c < 7.0%. Group B followed a conventional control strategy with a target HbA1c < 7.5%. Glycemic control in both groups was pursued through individualized adjustments to antidiabetic therapy, lifestyle modification, and monthly clinical follow-up.

Outcome measures and assessments

Assessments were performed at baseline and at the six-month follow-up.

Glycemic indices

Fasting plasma glucose (FPG), 2-hour postprandial glucose (2hPPG), and HbA1c were measured in local laboratory using standardized protocols. For FPG, two milliliters of venous blood were collected in fluoride-containing tubes after an overnight fast of 8–10 hours. For 2hPPG, two milliliters of venous blood were collected in fluoride-containing tubes exactly two hours after meal initiation. For HbA1c, whole blood was collected in fluoridecontaining, K_2-EDTA, or ammonium heparinized tubes. Before analysis, 25 µL of whole blood or reconstituted control was added to 1000 µL of hemoglobin denaturant (1:41 dilution), mixed gently to avoid foaming, and incubated at room temperature for five minutes.

Renal function parameters

Renal assessment included serum creatinine, blood urea, urine albumin-to-creatinine ratio (Alb/Cr), and eGFR.

Estimated glomerular filtration rate was calculated using the four-variable MDRD equation [12]; the race coefficient was not applied, as all participants were Egyptian.

Serum creatinine was measured in serum or heparinized plasma samples free of hemolysis and separated from red blood cells immediately after collection. Blood urea was measured in serum samples free of hemolysis. For Alb/Cr ratio, urine creatinine was measured in specimens collected in clean, leak-proof containers, and urine albumin was determined using either fresh spot urine samples or 24-hour urine collections. All measurements were performed using AU680 Beckman Coulter kits (Beckman Coulter Life Sciences, Inc.). eGFR (mL/min) was calculated using four-variable Modification of Diet in Renal Disease (MDRD) equation incorporating age, sex, serum creatinine, and race.

Neuropathy assessment

Neuropathy was assessed using DN4 questionnaire. DN4 includes four questions and a total of 10 items. Responses are recorded as “yes” or “no,” with “yes” scored as 1 and “no” as 0 to yield a total score out of 10. A score of ≥ 4/10 was considered positive for neuropathic pain (sensitivity 82.9% and specificity 89.9%) (Fig. 1).

Ophthalmologic assessment

Ophthalmologic assessment was performed using a dilated fundus examination with a non-mydriatic retinal camera (Optomed Aurora). Worsening of diabetic retinopathy was defined as progression of at least one step on Early Treatment Diabetic Retinopathy Study (ETDRS) severity scale, including progression from normal findings to non-proliferative diabetic retinopathy (NPDR), progression from mild to moderate or severe NPDR, or development of PDR [13].

Statistical analysis

Data handling and statistical analyses were performed using SPSS version 27 (IBM, Armonk, NY, USA). Continuous variables were described as mean ± standard deviation with observed range when distributions were approximately normal, and as median with interquartile range when distributions were non-normal. Categorical variables were summarized as counts and percentages. Group differences in categorical variables were tested using chi-square test, with Fisher's exact test applied when expected cell counts were < 5. For continuous variables, comparisons between two independent groups were conducted using independent-samples t test for normally distributed data and Mann–Whitney U test for nonnormally distributed data. All analyses were two-sided, and p-values < 0.05 were considered statistically significant.

Results

Age and sex distribution did not differ significantly between the two groups (age: p = 0.21; sex: p = 0.586). At baseline, HbA1c was significantly lower in Group A than Group B (9.02 ± 1.04% vs. 10.10 ± 1.28%, p < 0.001), and baseline urea was also lower in Group A (9.30 ± 3.37 vs. 11.57 ± 5.22 mg/dL, p = 0.031). After 6 months, Group A achieved significantly lower fasting blood glucose (117.15 ± 20.77 vs. 138.09 ± 19.91 mg/dL, p < 0.001), lower 2-hour postprandial glucose (155.03 ± 25.49 vs. 176.91 ± 18.34 mg/dL, p <0.001), and lower HbA1c (6.23 ± 0.35% vs. 7.27 ± 0.23%, p < 0.001) than Group B.

Fundus examination findings were comparable between the two groups at baseline, with no significant difference in the distribution of normal findings and diabetic retinopathy (p = 0.920). After 6 months, fundus findings differed significantly between groups (p = 0.005), with Group A showing a lower proportion of normal examinations (24.2% vs. 55.3%) and a higher proportion of diabetic retinopathy (75.8% vs. 44.7%) compared with Group B.

Fundus examination findings at baseline and after 6 months between the studied groups

	Group A (n = 33)	**Group B (n = 47)**	**p-value**
Baseline			
Normal; n (%)	20 (60.6)	29 (61.7)	0.920
DR; n (%)	13 (39.4)	18 (38.3)	
After 6 months			
Normal; n (%)	8 (24.2)	26 (55.3)	**0.005***
DR; n (%)	25 (75.8)	21 (44.7)	

*significant p-value

DR — diabetic retinopathy

Microvascular complications were significantly more frequent in Group A than Group B (84.8% vs. 31.9%, $p < 0.001$), including higher rates of albuminuria (36.4% vs. 14.9%, $p = 0.026$), retinopathy progression (63.6% vs. 14.9%, $p < 0.001$), and neuropathy (39.4% vs. 10.6%, $p = 0.002$).

Discussion

Type 2 diabetes is a chronic metabolic disorder characterized by persistent hyperglycemia and insulin resistance. These abnormalities contribute to atherosclerosis and endothelial dysfunction and increase microvascular complications risks, including nephropathy, neuropathy, and retinopathy [14]. Glycemic control, commonly assessed by HbA1c, remains central to diabetes management; however, optimal intensity and pace of HbA1c reduction remain debated, particularly with respect to microvascular outcomes [15].

Long-term data from UK Prospective Diabetes Study (UKPDS) showed that early and sustained improvements in glycemic control provide lasting protection against microvascular complications, a phenomenon known as "legacy effect." Importantly, these benefits were associated with gradual rather than rapid reductions in HbA1c, underscoring the relevance of the rate of glycemic improvement [6].

Our short-term findings should be interpreted within this time-horizon framework: early retinal responses over months may differ from the net benefits of sustained glycemic improvement over years. In patients with established retinopathy, rapid and/or large glycemic improvement can precipitate "early worsening," whereas longer-term maintenance of improved glycemic exposure is generally protective. Thus, our results do not refute the longterm microvascular benefit observed in UKPDS and other trials; rather, they highlight a clinically important early vulnerability during treatment intensification in individuals with pre-existing NPDR. Early worsening may be transient in some patients and potentially mitigated by a more gradual HbA1c reduction strategy, avoidance of abrupt glycemic shifts when feasible, and proactive ophthalmologic surveillance during intensification.

This randomized controlled trial evaluated the effects of tight versus less tight glycemic control on microvascular complications progression in 80 Egyptian patients with poorly controlled T2D. Although Group A (tight control, HbA1c < 7.0%) achieved significantly lower HbA1c, fasting plasma glucose, and postprandial glucose levels than Group B (HbA1c < 7.5%), Group A paradoxically showed higher rates of diabetic retinopathy progression and a higher overall incidence of microvascular complications. Importantly, the excess in the composite outcome was driven predominantly by retinopathy progression, with smaller contributions from albuminuria and DN4-positive neuropathic pain, supporting retinopathy worsening as the principal short-term safety signal in this cohort.

This finding is consistent with prior report describing "early worsening" of diabetic retinopathy after rapid improvement in glycemic control [16]. Proposed mechanisms include osmotic changes within retinal microvasculature, increased vascular endothelial growth factor (VEGF) expression associated with insulin therapy, and heightened oxidative stress [17, 18].

Because HbA1c reflects average glycemia rather than the temporal pattern of change, it may not fully capture the pathophysiologic context of early worsening. Dynamic glycemic features — such as rapid excursions, glycemic variability, and the speed of improvement — may impose microvascular stress and help explain heterogeneity in early retinal responses despite improved HbA1c.

For example, DCCT reported an increased risk of retinopathy progression during first year of intensive therapy despite long-term benefits [19]. Similarly, the ACCORD trial found that tight glycemic control (target HbA1c < 6.0%) reduced microalbuminuria but did not improve vision-related outcomes and was associated with increased mortality [20]. Consistent with this complexity, 84.8% of participants in tight-control group developed new microvascular complications or showed progression, compared with 31.9% in less tightcontrol group.

Renal outcomes showed a non-significant reduction in albuminuria in the tight-control group, whereas the less tight group exhibited a statistically significant improvement. This pattern is broadly consistent with findings from the ADVANCE trial, which reported a 65% reduction in the risk of ESRD with tight glycemic control; however, estimates had wide confidence intervals and subgroup results were not uniform.

Regarding diabetic neuropathy, we found no statistically significant difference between groups, although less tight-control group showed a modest clinical improvement. Similar findings were reported by Wang et al. [11], who observed a lower incidence of neuropathy over six years among patients with HbA1c < 7.5%. The ADA Standards of Care 2024 emphasize individualized HbA1c targets to balance microvascular benefits against risks of hypoglycemia and early worsening, particularly in patients with established retinopathy [10]. Together, these findings support our conclusion that rapid HbA1c reduction may not be appropriate for all patients. In addition, because DN4 primarily captures neuropathic pain, it may underestimate painless or subclinical neuropathy; therefore, neuropathy-related findings should be interpreted as symptomatic neuropathic pain outcomes rather than comprehensive peripheral neuropathy assessment.

Overall, these findings suggest that tight glycemic control should be implemented cautiously, particularly in patients with established microvascular disease. Both rate and magnitude of HbA1c reduction appear to be clinically important, as rapid declines may provoke acute microvascular stress, as observed in this and other studies [21]. Accordingly, a gradual, individualized glycemic target may be safer and more effective for reducing longterm complications.

From a clinical decision-making perspective, our data argue for focusing not only on “what HbA1c target” is selected, but also on “how” that target is achieved: staged intensification where feasible, attention to rapid glycemic shifts, coordinated care with ophthalmology, and scheduled retinal reassessment within 3–6 months after major therapy escalation in patients with NPDR.

These findings support a cautious, individualized approach to intensifying glycemic control, particularly in patients with established retinopathy. Rapid HbA1c reductions should be accompanied by closer ophthalmologic surveillance, with repeat retinal examination within 3–6 months of treatment intensification to detect early worsening. Gradual titration of therapy, patient education regarding new visual symptoms, and coordinated follow-up between diabetologists and ophthalmologists may further reduce short-term retinal risk.

Further research is needed to determine optimal rate of HbA1c reduction that maximizes metabolic benefit while maintaining safety. A prospective two-year extension study assessing retinal, renal, and neuropathic outcomes would clarify whether early worsening observed here is transient or sustained. Larger multicenter trials incorporating different glucose-lowering strategies and retinal imaging biomarkers, such as optical coherence tomography angiography (OCTA), would also help refine evidence-based glycemic targets for patients with established microvascular disease.

Building on the observed dissociation between improved HbA1c and short-term microvascular risk, future diabetes care may increasingly incorporate continuous glucose monitoring (CGM)-derived dynamic markers that capture the rate, variability, and temporal context of glycemic change rather than relying on static HbA1c thresholds alone [22]. Recent machine-learning studies using CGM profiles have also demonstrated the feasibility of predicting diabetic retinopathy risk, underscoring the potential of data-driven approaches to identify individuals vulnerable to early microvascular worsening during treatment intensification [23]. Integrating CGM-based temporal features with clinical variables into practical workflows could support earlier risk stratification, safer glycemic optimization, and more personalized, patient-centered management in high-risk T2D populations [22].

This study has limitations. Sample size estimation relied on DCCT retinopathy progression rates from type 1 diabetes, which may not reflect event rates in T2D; therefore, the trial was powered for retinopathy progression but may be underpowered for secondary renal and neuropathy outcomes. Because simple, unstratified randomization was used, baseline HbA1c differed between groups, introducing potential confounding despite adjustment/sensitivity analyses. Follow-up was limited to 6 months, capturing early changes (including early retinopathy worsening) without assessing longer-term stabilization or benefit. Neuropathy was assessed using the DN4 questionnaire, which primarily detects symptomatic neuropathic pain and may underestimate painless/subclinical neuropathy. Finally, all participants had mild–moderate NPDR at baseline, so findings relate to progression and may not generalize to patients without pre-existing retinopathy.

Conclusions

Although achieving lower HbA1c improved glycemic indices, it was associated with substantially higher short-term retinopathy worsening and greater overall microvascular deterioration in this high-risk cohort. Clinical implementation should therefore prioritize ratemodulated glycemic intensification — gradual HbA1c reduction and avoidance of abrupt glycemic shifts where feasible — together with individualized targets and scheduled retinal surveillance during treatment escalation, rather than target-based control alone.